Ayush Kumar Maurya

B.Tech. (Hons.) Chemical Engineering + M.Tech. Financial Engineering (5-Year Dual Degree)
Micro Specialisation in Artificial Intelligence & Applications | Indian Institute of Technology Kharagpur
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EDUCATION

Year	Degree / Exam	Institute	CGPA / Marks
2028	Dual Degree (5Y), Chemical & Financial Engineering	IIT Kharagpur	8.93 / 10
2022	Class XII, ISC	City Montessori School, Lucknow	97.0%
2020	Class X, ICSE	City Montessori School, Lucknow	96.8%

TECHNICAL SKILLS

Languages	Python, C++, C, Java, JavaScript, TypeScript, SQL, R, HTML
Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, HuggingFace Transformers, PEFT, NLTK, SHAP, DiCE
GenAI & Agents	LangChain, LangGraph, RAG, FAISS, Qdrant, OpenAI, Gemini, Groq, Llama, n8n
Quant & Statistics	Time Series (ARIMA, SARIMA, GARCH), Monte Carlo Simulation, Portfolio Theory, Option Pricing, Econometrics (Double ML, FGLS, Cochrane-Orcutt), DCF Valuation
Backend & Data	Node.js, Express, FastAPI, React, MongoDB, PostgreSQL, Drizzle ORM, Redis, BullMQ, Supabase, Databricks, Superset
Tools	Git, GitHub, GoogleTest, Streamlit, Selenium, BeautifulSoup, Jupyter, Google Colab, MATLAB, Excel

EXPERIENCE

University of Oulu, Finland — **May'26 – Present**
Research Intern — *Guide: Prof. Shuai Li*

- Built **9 MongoDB aggregation pipelines** for a robot interaction dashboard on latency trends, confidence and rankings
- Implemented a **Node.js/MongoDB backend** and a Gaussian Python simulator (mean confidence **0.80**, latency **800 ms**)
- Architecting a **React/Recharts** dashboard with an SSE feed and **AdaBoost/XGBoost anomaly detection** on session vectors

Zepto, Bangalore — **Apr'26 – Jun'26**
Procurement Analytics Intern — *Procurement Centre of Excellence*

- Realised **INR 6 Cr** in procurement cost savings by advising the CFO to scale reverse auctions, backed by bid analysis of **30+** past events
- Sped up reporting by **20%** via **Databricks** and **Superset** pipelines tracking global spend, PO/PR workflows and vendor performance
- Built weekly CFO dashboards for hard and soft savings while validating external workforce requests and vendor procurement data

Dr B C Roy Multi-Speciality Medical Research Centre, IIT Kharagpur — **Apr'26 – Jun'26**
Research Intern — *Guide: Dr. Mrinal Kumar Acharya*

- Built a **multi-agent FastAPI** AI dietician (Pydantic v2) computing BMR/TDEE baselines and generating **7-day Indian meal plans**
- Developed an **ICMR-NIN RAG pipeline** using **LangChain** and **FAISS** to ground agent outputs in national dietary safety guidelines
- Orchestrated task-specific LLM routing across **Llama-3.3-70B**, **Gemini-2.5-Flash** and **BioMistral** for real-time adherence analytics

ixana.ai, Bangalore — **Aug'25 – Jan'26**
AI Automation Engineer — *6 AI workflows, 7+ API integrations*

- **AI Project Manager:** Cut RCA reporting time by **90%** (**8.3 hrs/week**) with an **n8n** agent on ClickUp & MS Teams for **34 users**
- Automated **12-sheet** management reports and built nudging, resource allocator and Scrum Master agents for task updates
- **Procurement Agent:** Shortened procurement cycles by **50%** across **10+ vendors** by automating quotation parsing and PO drafting
- **HR Policy Assistant:** Built a **RAG** assistant on India/US HR policies using semantic chunking, OpenAI embeddings and **Qdrant**
- **Calendar Sync Engine:** Built a bi-directional Google-Outlook sync on n8n using **Supabase** snapshots to avoid duplicates

SG Analytics (acquired by Straive) — **May'25 – Jul'25**
Capital Markets Intern — *Investment Research*

- Supported bond investment decisions for a Swiss bank with **$190B+ AUM** by developing **70 credit profile factsheets**
- Analysed **25 sovereigns** on GDP, fiscal and monetary policy, and **45 financial institutions** on capitalisation, liquidity and asset quality
- Developed a macro-outlook report for a German credit rating agency, assessing growth forecasts and U.S. tariff impacts for **8 countries**

LongStraw Carbon — **Aug'26 – Present**
Founder's Office Intern — *Carbon Removal*

- Built a **3-scenario financial model** comparing equipment purchase vs leasing, projecting a **1.5-yr** breakeven and **43%** base-case IRR
- Created a **$250K+** revenue pipeline by identifying CDR buyers, drafting collateral, **5 RFP responses** and **3** BlueLayer storefronts

With/Care Ventures — **Oct'25 – Feb'26**
Founder's Office Intern (Founding Hire) — *Research & Foresight*

- Built an **AI intelligence tool** consolidating **150+** trend data points across **12 sectors** for AI-driven signal mapping and foresight
- Authored **6 foresight articles** on the digital economy, public policy and AI value chains, applying **8** foresight frameworks

360 Realtors LLP, Gurugram — **Jun'25 – Aug'25**
Founder's Office Intern — *Legal & Architecture Divisions*

- Steered an **INR 47L** procurement decision by auditing land management software and drafting feasibility reports
- Cut plotting time to **10 sec/plot** with a Google Maps-based platform automating village mapping of **800-1,000 plots** for 8 architects
- Automated GIS pipelines via custom **KML generation** and centroid computation, replacing manual tracking of land records

PROJECTS

Software Engineering

High-Performance Limit Order Book | *C++, AVL Trees, Hash Maps* *Jan'26*

- Engineered a **C++ matching engine** at **1.4M+ TPS** and **713 ns** average latency for Market, Limit, Stop and Stop-Limit orders
- Built AVL-balanced price levels with doubly-linked queues and hash maps for **O(1)** cancel/execute/lookup and **O(log M)** insert
- Developed a **5M-order** Gaussian generator and validated all order types through **GoogleTest** unit and integration tests

AI-Powered Email Client | *TypeScript, Microservices* *Nov'25 – Dec'25*

- Orchestrated **4 microservices** (Gateway, User, Email, AI) on Express with **JWT**-secured inter-service auth
- Processed up to **500 emails/hour** with centralised Gmail access via **Google OAuth 2.0**, **PostgreSQL** and **Drizzle ORM**
- Saved users **12 min/day** by automating summarisation and intent detection with the **Gemini API**, queued on **BullMQ & Redis**

Metro Route Planner | *C++, Graph Algorithms* *Jun'26*

- Built a C++ CLI app modelling **50 stations**, using **BFS** for unweighted and **Dijkstra's** algorithm for weighted shortest paths
- Structured a modular design separating graph logic and data, returning path, travel time and stop count in under **100 ms**

Waste Collection & Disposal Route Optimisation | *Self Project* *Mar'26 – Apr'26*

- Benchmarked optimised routes against a greedy baseline, achieving **7.5%** cost cut on Kaggle data and **23%** on a 42-point set
- Built a sensitivity module to stress-test fleet scalability across demand scenarios, exporting route maps and CSV artifacts

Quantitative Finance & Econometrics

GARCH-Informed Neural Networks for Option Pricing | *Quantitative Finance* *Jan'25 – Feb'25*

- Benchmarked MLP, LSTM and GARCH volatility models on call options, cutting mean pricing error vs the **Black-Scholes** baseline
- Achieved **0.048 RMSE** with an LSTM-based pricing model that outperformed MLP and GRU on NYSE options data

Sovereign Default Contagion with Neural Networks | *Financial Risk Management* *Feb'25 – Mar'25*

- Developed a **dual-encoder Transformer** on daily CDS spreads across **10 EU countries**, achieving a log-likelihood of **0.704**
- Identified **8 contagion patterns** during the European debt crisis using **Bayesian optimisation** for market segmentation

Price Elasticity Estimation with Double Machine Learning | *Prof. Adway Mitra* *Feb'26 – Apr'26*

- Estimated price elasticity of demand for **4.4K+** Steam games via **Double ML**, tackling endogeneity with NLP-based controls
- Engineered **2.7K+** TF-IDF and DistilBERT features, reducing omitted-variable bias by **21%** vs OLS through 3-fold cross-fitting

Monte Carlo Portfolio Forecasting & Risk Budgeting | *Self Project* *Jun'25 – Jul'25*

- Simulated **15K+** return paths on **6 NSE assets** using log-normal dynamics and **Ledoit-Wolf** covariance shrinkage
- Constructed the **Markowitz** efficient frontier to maximise Sharpe ratio, deriving an optimal equity-gold portfolio at **5% VaR**

Equity Research: Balkrishna Industries (NSE: BALKRISIND) | *Bronze, NMIMS Moolyankan 2024* *Sep'24 – Oct'24*

- Identified **31%** overvaluation via fundamental, relative and **DCF** valuation, issuing a target price of INR 2,314 with a sell call
- Stress-tested valuation for rubber price sensitivity at **8.84%** WACC, 24-28% EBITDA margins and 7-12% revenue growth

Inflation Modelling & Autocorrelation Detection | *Prof. Krittika Bandyopadhyay* *Mar'25 – Apr'25*

- Modelled **30 years** of India's inflation drivers (R^2 = **0.61**), confirming no autocorrelation via Durbin-Watson and Runs tests
- Improved model robustness by **25%** with **Cochrane-Orcutt** and **FGLS**, validating coefficient stability under outliers

Machine Learning & AI

Credit Card Fraud Detection | *ULB Dataset, Self Project* *Mar'26 – May'26*

- Benchmarked SMOTE, class weighting and **CTGAN** with LR, RF and XGBoost at **0.17%** fraud rate (492/284K), using leakage-free PR-AUC
- Achieved **0.92 precision** and **0.84 F1** with CTGAN + XGBoost, outperforming SMOTE-based oversampling
- Found that transaction amounts of 0, 1 and 99.99 made up **33.94%** of all fraud cases, revealing key behavioural patterns

SMS Spam Classification: Classical NLP to Transformers | *Self Project* *May'26*

- Benchmarked BoW, TF-IDF, Word2Vec, GloVe, Bi-LSTM and **DistilBERT** with LR, NB, RF and SVM on **5,169** SMSs (13% spam)
- Achieved F1-macro of **0.982** with Bi-LSTM, ahead of DistilBERT (**0.981**) and TF-IDF + Linear SVM (**0.971**)

Generative Image Augmentation for CIFAR-10 | *Self Project* *Jun'26*

- Engineered a **DCGAN** (ConvTranspose generator, strided-CNN discriminator) to augment a CNN classifier
- Generated realistic samples at **IS 5.2** and **FID 48**, in line with baseline DCGAN performance on CIFAR-10

Regression & Time Series Forecasting | *Course Term Project* *Apr'26*

- Benchmarked **8** regression and time-series models on 10 years of daily temperature data with lag features, ADF, ACF/PACF and VIF
- Achieved **RMSE 2.279** and **MAE 1.764** with Polynomial Regression, outperforming OLS, Lasso, ARIMA and SARIMA

COMPETITIONS

UCO Bank PSB Hackathon 2026 | *Finalist* *Apr'26 – Present*

- Designed **SAKHI**, an alternative credit scoring framework for bureau-invisible borrowers built on UPI transaction data
- Generated **25K** synthetic UPI profiles via probabilistic simulation and trained a **LightGBM** default-risk model (AUC **0.674**)
- Added psychometric verification and a **LangChain-Groq** chatbot using **SHAP** and **DiCE** to explain scores and suggest actions

Bharatiya Antariksh Hackathon 2026 | *ISRO* *Jun'26 – Present*

- Fused SoLEXS and HEL1OS X-ray streams, processing **10M+** observations for real-time solar flare and space-weather alerts
- Engineered **24+** temporal features and benchmarked LightGBM, XGBoost and LSTM with physics-informed losses
- Achieved **ROC-AUC 0.81**, **85%** recall, **<15%** false alarm rate and **10-min** early warning with SHAP-based explainability

Prodylitics 2024, IIM Indore | *Gold Winner among 1,000+ teams* *Feb'25 – Apr'25*

- Analysed **1,240** orders to find drivers of customer ratings for a food delivery platform, proposing 6 strategic solutions
- Projected **70%** NPS growth and a **30%** market share capture ($9.2bn SOM), validated through **90+** primary surveys

General Championship Case Study Challenge 2024-25 | *Vice Captain, Rajendra Prasad Hall* *Mar'25 – Apr'25*

- Built a multi-parameter Empathy-Resilience assessment model for **Deloitte** with **15+** variables, combining survey data and NLP
- Applied LSTM and transformer-based models to predict employee outcomes, achieving **78%** accuracy and R^2 = **0.995**

AWARDS & ACHIEVEMENTS

- Ranked in the **top 2%** of **1,900+** students to secure a department change and a switch to the Financial Engineering dual degree
- Ranked **112th** among **13K** participants in HackerRank Orchestrate June'26, a 24-hour agentic AI hackathon
- **Gold Winner** at Prodylitics 2024 (IIM Indore) and **Bronze Winner** at NMIMS Moolyankan 2024 (equity research)

POSITIONS OF RESPONSIBILITY

Founding Team Member & Senior Executive **Jul'25 – Present**
Innovation and Incubation Cell *IIT Kharagpur*

- Organised **VibeBuild**, IIT Kharagpur's 36-hour AI hackathon, in partnership with MNFIC to drive campus-wide tech innovation
- Accelerated **30+** early-stage ventures through flagship initiatives *Chai with Titans* and *Entrepreneurs First*
- Forged alliances with **5** premier institutes (IIM-A, IIM-L, IIM-I, IIT-B, IIT-M) and onboarded CXOs as mentors

Head of Sponsorship **Jun'25 – Present**
Chemical Engineering Association *IIT Kharagpur*

- Generated **INR 5L+** in funding for Fugacity 2025 through sponsorship deals with IOCL, Aspen, COMSOL and Unstop
- Organised alumni talks with senior industry leaders for **200+** students; drove the ChE Innovation Centre proposal

Associate Manager **Jun'24 – Jun'25**
Entrepreneurship Cell *IIT Kharagpur*

- Secured **INR 58L+** in sponsorships for GES 2025, leading hospitality for **50+ CXOs**, **1,500+** attendees and **20+** partners
- Led EAD Kanpur 2024 at CSJMU, securing **INR 22L+** in sponsorships and media coverage across **10+** local outlets

EXTRACURRICULAR ACTIVITIES

- **General Championship:** Represented Rajendra Prasad Hall in Product Management, Cricket and Chess in Inter-Hall GC 2025-26
- **Volunteering:** Taught **55** primary students, organised a health camp for **100+** patients and led 4 Think India initiatives (NSS)
- **Illumination:** Secured **2nd place** among **20+** halls at Inter-Hall Illumination, leading concept, design and execution